

AI Chatbot (Customer Support Virtual Agent using Natural Language Processing)

Presented by

23BAI11114 MALISHKA SALIL PARADKAR

23BAI11146 SAHU SAMIKSHA SAMEER

23BAI11147 DIVYANSH BHARDWAJ

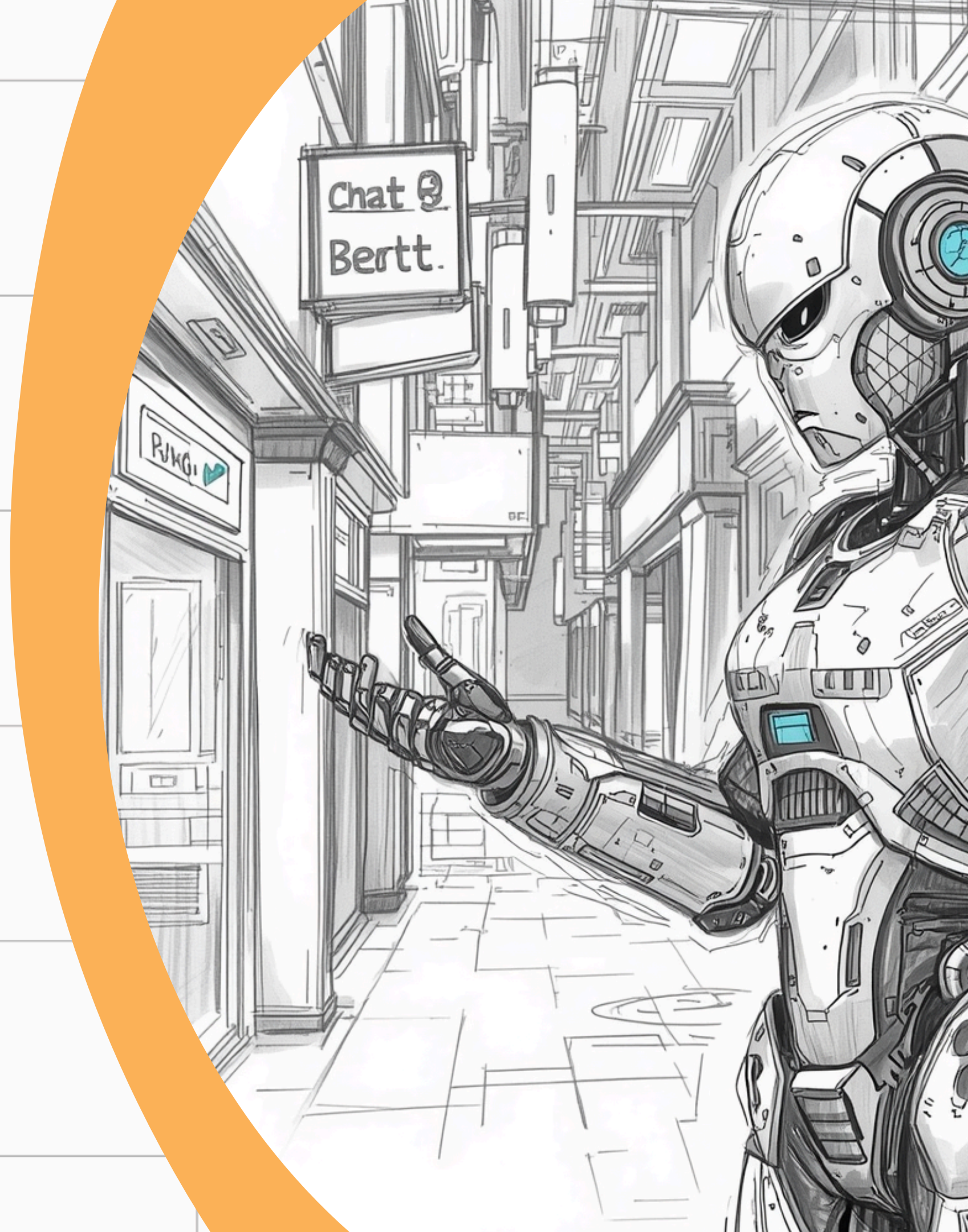
23BAI11198 ARYA KHULLAR

23BAI11205 AASTHA NAGAR

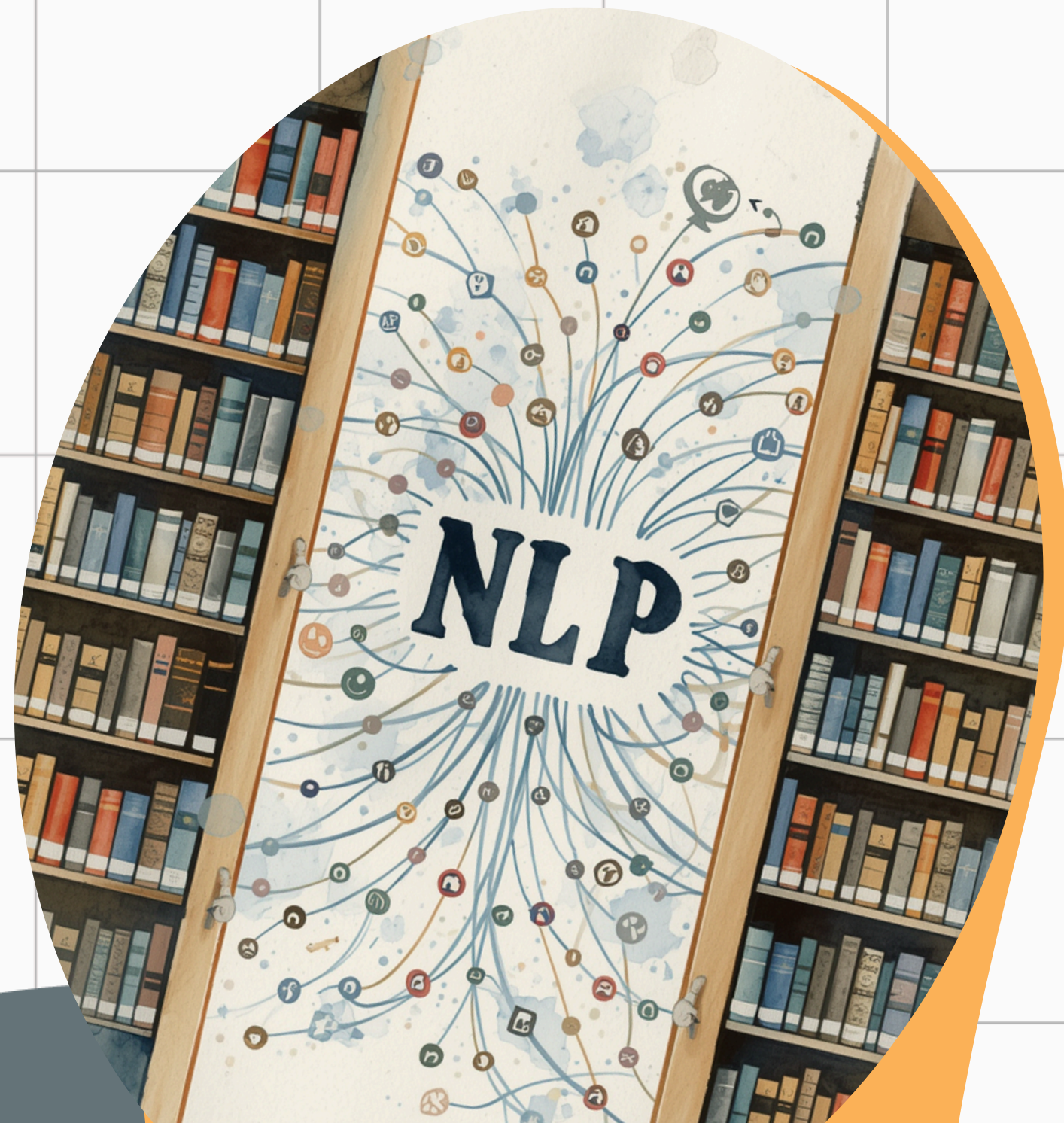
23BAI11209 GARV GUPTA

Introduction – What is an AI Chatbot?

- An AI Chatbot is a software that can interact with users in natural language.
- It simulates human conversation using text or voice.
- Uses Natural Language Processing (NLP) to understand and respond.
- Can handle multiple queries instantly without human intervention.



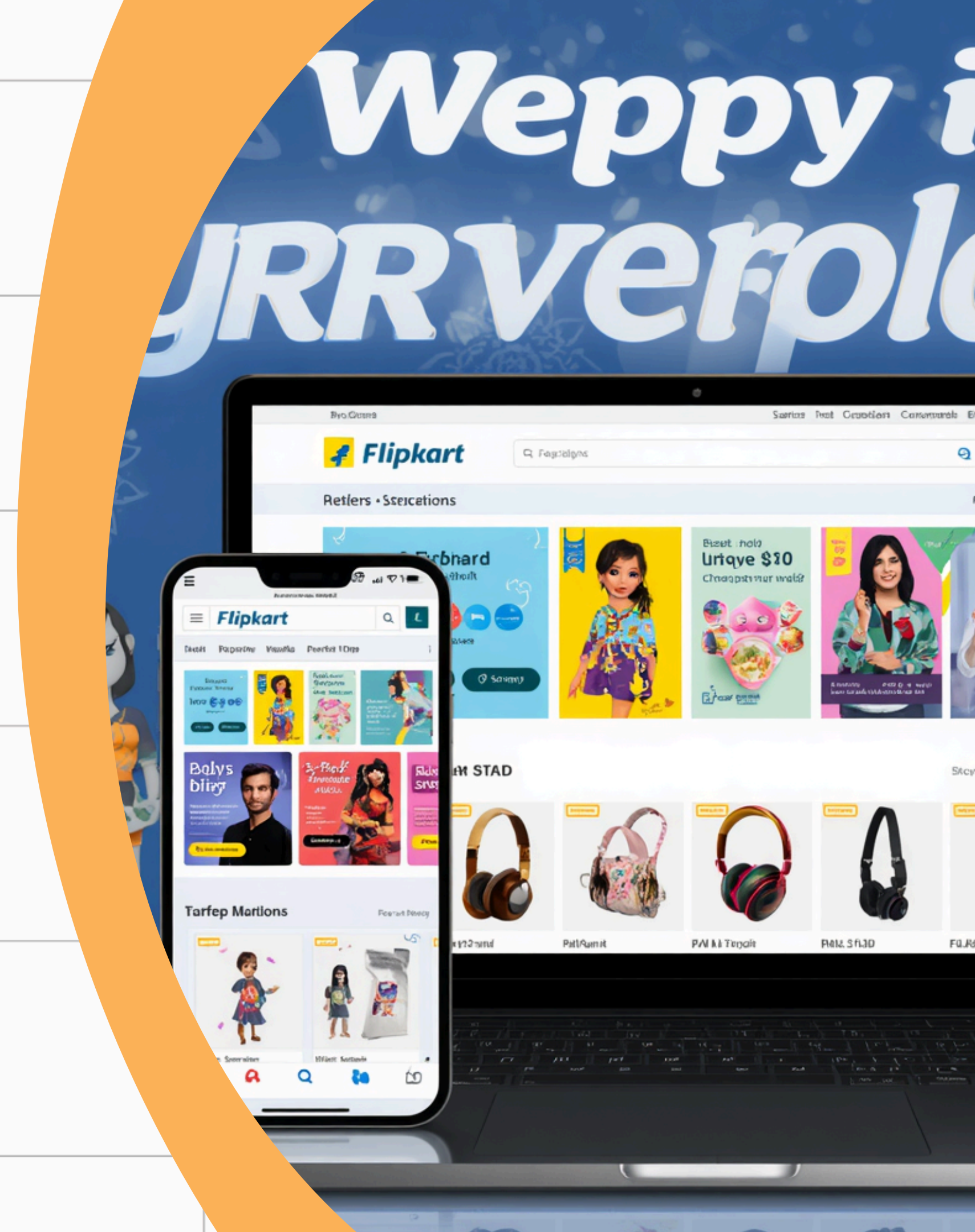
Role of NLP in Chatbots



- NLP helps chatbots understand human language.
- Converts user input into structured data.
- Key tasks:
 - Text understanding
 - Intent detection
 - Entity extraction
- Enables meaningful and context-aware responses.

Real-World Use of Chatbots

- Widely used in customer support systems
- Examples:
- Order tracking
- Complaint handling
- FAQs automation
- Used by companies like:
- Amazon
- Flipkart
- Banking apps



Case Study Overview



- Example: Customer Support Chatbot
- Handles:
 - Order tracking
 - Complaints
 - General queries
- Works 24/7 without human involvement
- Reduces workload on human agents

Example User Input



- User Input:
- “I want to track my order 123”
- What chatbot does:
- Understands request type → Track order
- Extracts order ID → 123
- Gives response → Order status

NLP Pipeline in AI Chatbot



1. Data Preprocessing:

- a. Sentence Segmentation
- b. Word Tokenization
- c. Stopword removal
- d. Normalization

2. Feature Extraction

Converting text into numerical features by assigning importance to words using TF-IDF values.

3. Modelling

Applying machine learning algorithms (e.g., Naive Bayes) to classify user intent.

How Chatbot Understands User Input

Input:
"I want to track my order 123"

↓

Preprocessing
(track, order, 123)

↓

Feature Extraction
(TF-IDF → important words)

↓

Modelling
(Intent: Track Order)

↓

Response:
"Your order 123 is out for delivery"

Tokenization → [I, want, to, track, my, order, 123]
Stopword Removal → [want, track, order, 123]
Normalization → [track, order, 123]

Word	TF	IDF	TF-IDF
track	0.33	0.3	0.1
order	0.33	0.1	0.03
123	0.33	1	0.33

IDF values are derived from the corpus and given in simplified form for calculation.

Regular Expressions (Regex):

Pattern matching technique used to extract specific data from text

In Chatbots, used to identify structured information

Example:

Input: "Track my order 123"

Pattern: **[0-9]+**

Output: 123 (Order ID)



Input → Regex → Output

"Track my order 123" → [0-9]+ → 123

Uses:

Extract Order ID

Detect phone numbers, emails, etc.

Regex is efficient for extracting fixed patterns without using machine learning.

Finite State Automata (FSA)

Definition:

A mathematical model consisting of states and transitions
Used to represent step-by-step processing

In Chatbot:

Controls the conversation flow

Moves from one state to another based on user input

"NLP pipeline answers what the user wants, while FSA decides what the system should do next."

[Start State] → [Query State] → [Processing State] → [Response State]

Case Study:

- Start State: waits for input
- Query State: receives "I want to track my order 123"
- Processing State: NLP applied → Tokenization, TF-IDF
 - Intent: Track Order
 - Entity: 123
- Response State: "Your order 123 is out for delivery"

Understanding Word Structure: Morphology

Before the chatbot can understand the 'meaning' of a sentence, it needs to understand the 'structure' of the individual words. This is where Morphology comes in. users use different forms of the same word.

Morphological Analysis



Breaking down words into their root form (Morphemes).

Stemming & Lemmatization:



to strip away suffixes and prefixes.
Example: "Tracking" or "Tracked" → Track.

Inflection & Derivation:



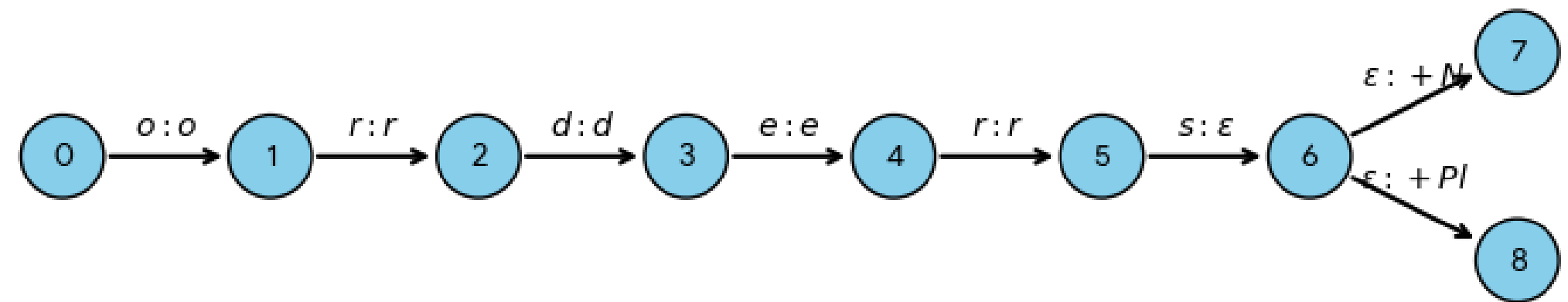
How "Order" becomes "Ordering" (inflection) or "Orderly" (derivation).

Finite State Transducers (FST):

It is an extension of FSA that maps input strings to output strings. It consists of states, input symbols, and output symbol and used for morphological analysis (understanding word structure)

Chatbot analyzes "surface words" into root forms and grammatical features. It moves beyond recognition to transformation.

[Surface Word: "orders"] → [Root: "order" + Tag: "Noun" + Tag: "Plural"]



FSA recognizes if a word is valid, while FST translates what that word actually represents.

Case Study: Word Analysis

Input Word: "orders"

Step-by-Step Processing:

State q₀: Reads "o-r-d-e-r" → Maps to root "order"

Transition: Reads "s" → Triggers mapping to "Plural"

Final State: Outputs structured data for the chatbot to use

Lexical Result: order + N + Pl

Syntax Parsing

01. POS Tagging:

Assigning Parts of Speech (Noun, Verb, Determiner).

I → (PRON) want → (VERB) to → (PER) track → (VERB) my → (PRON) order → (NOUN).

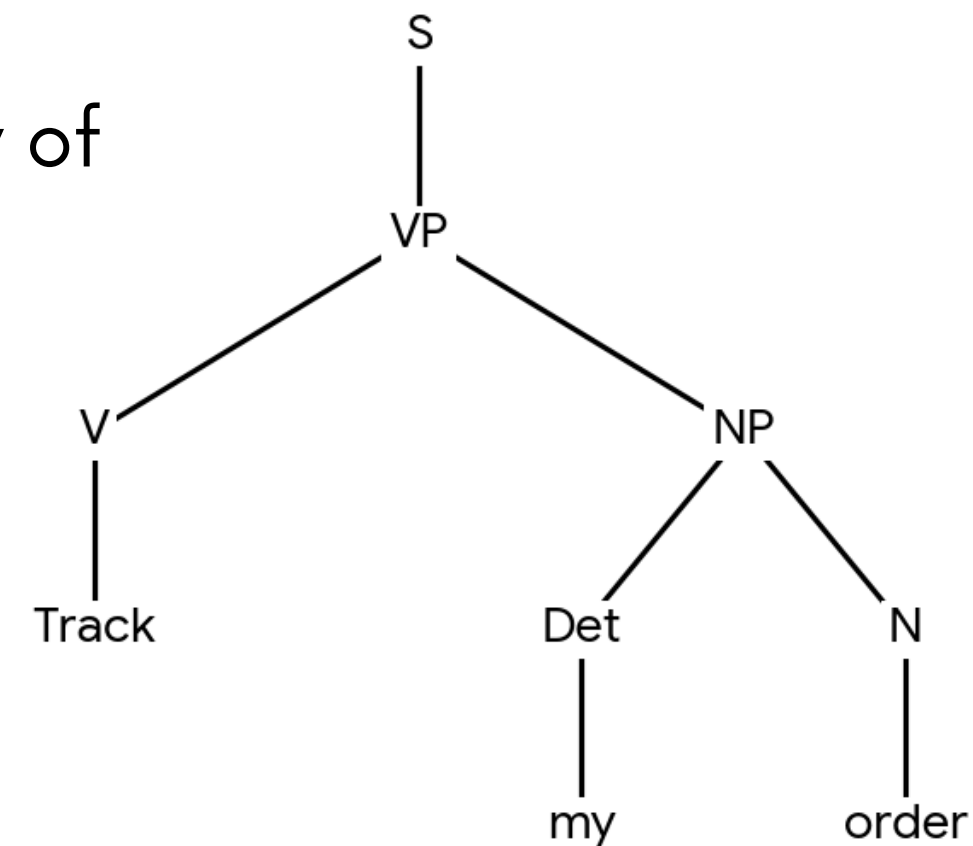
02. Context-Free Grammar (CFG):

A set of rules to define valid sentences.

Rule: $S \rightarrow NP + VP$ (Sentence consists of Noun Phrase + Verb Phrase)

03. The Parse Tree:

Visualizing the hierarchy of a sentence.



Semantic Analysis in Chatbots

- “Semantics is the stage where the chatbot actually understands what the user means.
- Focuses on understanding the meaning of user input, not just words
- Converts natural language into a structured representation

For example, if a user says ‘I want to track my order 123’, the chatbot doesn’t just read words—it extracts meaning:

- The user’s intent is tracking
- The object is order 123

FOPL Representation:

Request(User, Track(Order123))

Key Benefits:

Identifies user goal accurately

Reduces ambiguity in interpretation

Enables precise and relevant responses

Machine Learning in Chatbots

- “Machine Learning helps the chatbot automatically understand user intent based on data.
- Used for Intent Classification
- Converts text → predicted intent

Algorithms:

1. Naive Bayes :

Probabilistic classifier
Works well for text data

2. Decision Tree:

Rule-based classification
Easy to interpret

Text Normalization:

- Converts text into a standard format
- Techniques include:
 - Lowercasing
 - Removing punctuation

Example:

“TRACK Order!!” → “track order”

Where is my order → “track order”

Application

ML Techniques in Chatbots

Naive Bayes: Intent classification using probabilistic models

Decision Trees: Rule-based decision making

Text Normalization: Standardizing input variations

Real-World Importance

NLP transforms customer support by enabling chatbots to handle complex queries with human-like understanding, reducing response times from hours to seconds while maintaining accuracy.

Additional NLP Applications

Spell checking and correction

Language translation

Text summarization

Conclusion

NLP enables intelligent chatbots by allowing computers to understand and process human language effectively. This helps chatbots give accurate and human-like responses. In real life, NLP plays a crucial role in many applications such as customer support, translation, and voice assistants, making communication faster, easier, and more efficient.

- NLP helps chatbots understand human language and respond intelligently
- Makes conversations more natural and accurate
- Reduces human effort in customer support
- Used widely in real-world applications like translation, voice assistants, etc.